

Vessel Traffic Services (VTS)

Part 1 and Part 2

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 1, Exercise 1

What are vessel traffic services?

Growth **in** international commerce and tourism as well as technical development has resulted **in** high traffic density **in** many sea areas. Moreover, vessels are bigger and faster. The approaches **to** pilot boarding stations, ports and river waterways are all areas **of** potential traffic congestion and complexity. To reduce the risk **of** collision **in** busy port waters as well as **in** more open coastal waterways vessel traffic services (VTS) have been implemented. Vessel traffic services are shore-side information processing systems which range **from** the provision **of** simple information messages **to** ships, to extensive management **of** traffic **within** a port or waterway. Information messages can include position **of** other traffic, defects **in** aids to navigation or meteorological hazard warnings.

As stated **in** the IMO Resolution A.857(20), the efficiency **of** a VTS will depend **on** the reliability and continuity **of** communications and **on** the ability to provide good and unambiguous information. Furthermore, the quality **of** accident-prevention measures will depend **on** the system's capability **of** detecting a developing dangerous situation and **on** the ability to give timely warning **of** such dangers.

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 1, Exercise 2

1. B
2. A
3. B
4. A
5. B
6. B
7. B

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 2, Exercise 1

In the event of pollution, VTS directs other vessels to **<avoid>** the area.

Vessel traffic services **<improve>** the safety and efficiency of vessel traffic.

Vessel traffic services **<apply>** to all merchant and government vessels.

Leisure crafts, too, should **<follow>** the instructions given by the VTS centre.

VTS helps to **<limit>** the effects of pollution.

Also, in the event of pollution, VTS helps to **<identify>** the source of the pollution.

Leisure crafts may or may not **<participate>** in a VTS.

Vessel traffic services **<protect>** the marine environment.

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 2, Exercise 2

1. C
2. A
3. B
4. B
5. B

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 3, Exercise 1

identification	of	vessels
strategic planning	of	vessel movements
level	of	service
co-ordination	of	pollution response
provision	of	navigational information
prevention	of	pollution

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 4, Exercise 1

Responsibility for safe navigation lies with **<the ship's master>**.

The master of a vessel has knowledge of **<the vessel>**.

The main tool of a VTS operator is **<a traffic image>**.

The operators can monitor **<ships' movements>** online.

A VTS operator has specialised knowledge of **<the waterway>**.

An electronic **<navigational chart>** displays the fairway and aids to navigation.

<a VTS passage plan> cannot supersede the decisions of the master.

<a VTS operator> has responsibility for managing the traffic in the area.

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 4, Exercise 2

Responsibility for safe navigation <lies> with the ship's master.

A VTS operator has <responsibility> for managing the traffic in the area.

A traffic image is a <comprehensive> overview of the traffic in the area.

Ship's radar echoes are identified and put under <surveillance>.

An electronic navigational chart <displays> the fairway.

A VTS passage plan cannot <supersede> the decisions of the master.

First, the ship's movements are <monitored> by radar.

The operators can <predict> ship's future path

Then, the computer <combines> the radar image with the electronic chart.

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 5, Exercise 1

1. C
2. A
3. B
4. C
5. D
6. B
7. B

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 6, Exercise 1

an agent	a verb		a receiver	time	location
a vessel	submits	a report of deficiencies	to a VTS centre	prior to entering	a VTS area

an agent		a verb	an object	a receiver		time	location
a vessel heading	for the port of Kotka	gives	a traffic report	to the Kotka VTS centre	via VHF	when entering	the Kotka VTS area

an agent	a verb		an object		a receiver	an object
the VTS centre	confirms	that it has received	the traffic report	and provides	the vessel	with instructions

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 7, Exercise 1

Information may **<include>** intentions of other traffic.

<priority> is given to urgent information.

The **<aim>** of information service is to assist on-board navigational decision-making.

Information service **<ensures>** that necessary information becomes available in time.

Vessels in the VTS area **<are obliged>** to report to the VTS Centre.

A VTS Centre provides information about factors that may **<influence>** the vessel's transit.

Urgent information is of immediate **<concern>** to the vessels in the area.

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 8, Exercise 1

1. to assist
2. to monitor
3. provided
4. occasions
5. circumstances
6. rendered
7. deemed
8. stated
9. acknowledged
10. authorised

Vessel Traffic Services (VTS) Part 1: VTS in theory, Unit 9, Exercise 1

1. prevent the development of dangerous maritime traffic situations
2. provide for the safe and efficient movement of vessel traffic
3. the operational management of traffic
4. the forward planning of vessel movements
5. routes to be followed
6. times of high traffic density
7. affect the flow of other traffic
8. vessels from overtaking other ships
9. the right-of-way in narrow channels
10. include establishing and operating a system

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 1, Exercise 1

Kotka VTS, this is Marina.	Marina, Kotka VTS.
We are passing the reporting point number 10. My ETA at pilot station is 1300 hours local time.	Marina, Kotka VTS. You are in the Kotka VTS area. Proceed to Orrengrund Pilot Station.
Information: Rig the pilot ladder on starboard side. One metre above the water. Make a boarding speed of six knots.	Pilot ladder starboard side. One metre above the water. Boarding speed six knots.
Is pilot ready for me?	Yes, pilot on arrival.
Traffic information: One outbound vessel, name Annegrecht, now passing Tainio light.	Okay, traffic information received.
Information: There are cable operations in position 277 degrees from the southern point of Kaunissaari island, distance 4 miles. Wide berth requested.	Cable operations in vicinity of Kaunissaari. Wide berth requested. Well received.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 1, Exercise 2

- Kotka VTS, this is Marina.
- Marina, Kotka VTS.
- We are passing <-> the reporting point number 10. My ETA <at> pilot station is <-> 1300 hours local time.
- Marina, Kotka VTS. You are <in> the Kotka VTS area. Proceed <to> Orrengrund Pilot Station.
- Information: Rig the pilot ladder <on> starboard side, one metre <above> the water. Make a boarding speed <of> six knots.
- Pilot ladder <-> starboard side, one metre <above> the water. Boarding speed <of> six knots. Is pilot ready <for> me?
- Yes, pilot <on> arrival. Traffic information: One outbound vessel, name Annegrecht, now passing <-> Tainio light.
- Okay, traffic information received.
- Information: There are cable operations <in> position 277 degrees <from> the southern point of Kaunissaari island, distance 4 miles. Wide berth requested.
- Cable operations <in> vicinity <of> Kaunissaari. Wide berth requested. Well received.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 1, Exercise 3

- Kotka VTS, **this is** Marina.
 - Marina, Kotka VTS.
 - We are **passing the reporting point** number 10. My **ETA** at pilot station is 1300 hours local time.
 - Marina, Kotka VTS. You are in **the Kotka VTS area**. Proceed to Orrengrund Pilot Station.
- Information: Rig the pilot ladder on **starboard side**, one metre above the water. Make a **boarding speed** of six knots.

- Pilot ladder **starboard side**, one metre above the water. **Boarding speed/boarding speed** of six knots. Is pilot ready for me?
 - Yes, **pilot on arrival**. Traffic information: One outbound vessel, name Annegrecht, now **passing** Tainio light.
 - Okay, traffic information received.
- Information: There are **cable operations** in position 277 degrees from the southern point of Kaunissaari island, distance 4 miles. **Wide berth/wide berth** requested.
- **Cable operations/cable operations** in vicinity of Kaunissaari. **Wide berth/wide berth** requested. Well received.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 2, Exercise 1

- Marina, Kotka VTS.
 - Kotka VTS, Marina.
 - Marina, Kotka VTS. Information: The fairway **<is>** on the west side of Tainio racon.
 - Yes, I will **<alter>** course to port.
 - the ship is **<proceeding>** towards Tainio racon.
- Marina, Kotka VTS. Warning: You are **<running>** into danger. The fairway **<is>** on the west side of Tainio racon.
- Okay, I will **<leave>** it on my starboard side.
 - Information: One outbound vessel, name Finnmill, **<passing>** Orrengrund after 10 minutes.
 - Understood, Finnmill is **<coming>** out. I can **<see>** her on my radar screen.
 - Information: There is a yacht race on the west side of Kotka lighthouse. The yachts are **<heading>** north/west.
 - Well understood.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 2, Exercise 2

Vessel is

proceeding from harbour/anchorage to sea	outbound
proceeding across a fairway, lane or route	crossing
leaving a berth or anchorage	departing
making a large alteration of course	turning
proceeding from sea to harbour or dock	inbound

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 2, Exercise 3

Riga Traffic, **this is Iris calling/this is iris calling** on channel 09.

Iris, **Riga Traffic/riga traffic**

I'm entering **your area**, proceeding **inbound**.

Question/question Do you have any information about **outbound vessels**?

Answer/answer There are two **outbound vessels**. The first one is Durham, **two decimal five miles west of/2.5 miles West of** Daugavgriva lighthouse, **bound for** Liepaja, and the second one is Lumumba, **one mile behind/1 mile behind** Durham.

Well received, **two outbound vessels/2 outbound vessels**, the first one is Durham and the second one is Lumumba.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 3, Exercise 1

- 1) 035 5.5
- 2) 10
- 3) 1
- 4) 71
- 5) 0920
- 6) 4 10
- 7) 120 2.2
- 8) 240 0.2 6
- 9) 090
- 10) 140 1.8
- 11) 2
- 12) 0945

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 3, Exercise 2

- Hightower VTS, this is MSC Marianna.
- MSC Marianna, Hightower VTS.
- Information: We have problems with electricity on the bridge. Only one radar is working. I require navigational **assistance**.
- Understood. You have problems with electricity. Question: What is your position?
- Answer: My position is **bearing** 035 degrees, **distance** 5.5 miles from Landvall lighthouse.
- **Bearing** 035 degrees, **distance** 5.5 miles from Landvall lighthouse. I have located you on my **radar screen**.
- All **information** is based on VTS equipment. **Stand by** on channel 10. If you do not hear me in one minute time, navigational assistance is ended. In that case go back to channel 71 and call Hightower VTS. Navigational assistance starts at 0920 local time.
- Okay. Navigational assistance is starting and is provided on channel 10. If I do not hear from you at one minute intervals, navigational assistance is ended and I will call Hightower VTS on channel 71.
- All ships, Hightower VTS. Traffic information: Container vessel MSC Marianna, just passed buoy number four, is receiving navigational assistance on channel 10 and is not monitoring any other VTS working channel.
- ...
- You are on the centre of the fairway, tendency to north. **Bearing** to the next buoy is 120 degrees, **distance** 2.2 miles.
- Thank you. Information: My next waypoint is in position 240 degrees and 0.2 miles from buoy number six. After that I will commence the turn to the next **course** which is 090.
- Next waypoint in position 240 degrees and 0.2 miles from buoy number six. Understood.
- ...
- You are on the northern buoy-line of the fairway, tendency to north. **Bearing** to the next waypoint is 140 degrees, **distance** 1.8 miles.
- Understood. I am on the northern buoy-line. **Bearing** to the next waypoint 140 degrees. I will alter my **course** to south.
- ...
- I am passing buoy number two and my navigational equipment is working now. I now longer require navigational assistance. Thank you.
- MSC Marianna, Hightower VTS. Information received. Navigational assistance ends at 0945 local time.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 3, Exercise 3

Recommend you **keep** your present course.

Recommend you **alter** course to 125 degrees True.

Recommend you **make** course 345 degrees True.

You are **steering** a dangerous course.

Vessel Autosan ahead of you **is** on opposite course.

Vessel Peikko is **constrained** by draught.

Maximum **permitted** draught is 11 metres.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 1

Kotka VTS, this is Janra.

Janra, Kotka VTS.

Information: I am ready to depart from Halla and I am bound for Orrengrund.

Traffic clearance. Janra has permission to depart from Halla to Orrengrund at 0900 local time.

We have permission to depart.

Traffic information: One outbound vessel Ladoga-53 has just passed buoy number 2 and tanker Berg is anchored in position one mile south from Vehkaluoto.

Okay, thank you for your information. I can see Ladoga-53.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 2

Kotka VTS, Janra. We have 10 minutes to the strait of Ruotsinsalmi.

Janra, Kotka VTS. You have 10 minutes to the strait of Ruotsinsalmi. There is no other reported traffic in the strait.

Well received.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 3

Kotka VTS, this is Greta. I request clearance to depart from Sunila bound for Halla. I am just shifting berth.

Greta, Kotka VTS. Negative. Do not proceed.

Traffic information: Container vessel Janra has two minutes to the strait of Ruotsinsalmi. I will contact you when she has passed the strait and the fairway is free.

Well received. No permission to depart. I will remain alongside.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 4

depart	from	a berth
bound	for	a pilot station
shift	from	the berth number 2 to anchorage
remain	alongside	a berth
pass	-	the Strait of Ruotsinsalmi
enter	-	a fairway
leave	-	a fairway
block	-	a channel

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 5

- Margareta, this is Kotka VTS calling.

- Kotka VTS, Margareta.

- Margareta, Kotka VTS. **Information/information.** On your **east/East** side is Silva and she is **heading to Orregrund pilot boarding position.**

- Thank you for **your information.** I can see her on my **radar screen.** Question:

Who is **taking pilot** first? **Am I/am i** number one or is Silva **number one/number 1?**

- **Answer/answer:** Silva is **number one/number 1** and you are **number two/number 2.**

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 4, Exercise 6

Kotka VTS, this is Grimm.

Grimm, Kotka VTS.

Sailing plan: Orregrund, the strait of Ruotsinsalmi, Halla.

Sailing plan: Orregrund, the strait of Ruotsinsalmi, Halla.

Information: One outbound vessel Vera just passing Kukouri.

Well received. Vera passing Kukouri.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 5, Exercise 1

- Kotka VTS, this is Marina.
- Marina, Kotka VTS.
- Kotka VTS, Marina. We have **engine problems** and I am **manoeuvring with difficulty**.
- Marina, Kotka VTS. Do you need **assistance**?
- **Negative/negative**, no assistance needed.
- Understood. Keep me **updated** about your situation.
- Kotka VTS, this is Nordic Star.
- Nordic Star, Kotka VTS.
- Intention: Nordic Star is ready to depart from **berth** number C5 to sea. Can I have **clearance** now?
- Negative. Do not proceed. Traffic information: Tanker Paula is just departing from oil terminal.
Call VTS again after Paula has **passed** you.
- Well understood. We do not have **permission** to depart. We will **stay alongside**. I will call you again when tanker Paula has passed me.
- Friedrich Russ, Kotka VTS.
- Kotka VTS, this is Friedrich Russ.
- Information: Outbound vessel Marina **in the vicinity** of buoy number 10 has engine problems and she is **manoeuvring with difficulty**. Advice: Keep minimum passing distance of five **cables** when passing her.
- Well understood. I will **keep clear** of Marina.

- Kotka VTS, Marina.
- Marina, Kotka VTS.
- My engines are working now and I can **make a speed** of 5 knots. I am still **manoeuvring with difficulty**
- Understood. Traffic information: **Outbound vessel/outbound vessel** Friedrich Russ has just passed buoy number 8.
- Friedrich Russ has passed buoy number 8 outbound. Okay. I will keep to the east side of the **fairway**.
- Friedrich Russ, Kotka VTS.
- Kotka VTS, Friedrich Russ. I heard that Marina is keeping to the east side of the **fairway**. I am **passing** Marina on her west side maintaining the minimum **passing distance** of 5 cables.
- Friedrich Russ, Kotka VTS. Thank you.
- Kotka VTS, Marina.

- Marina, Kotka VTS.
- I still have problems with engines. Question: Where can I **anchor**?
- Question: What is your **maximum draught/maximum draft**?
- Answer. My **maximum draught/maximum draft** is 6.5 metres.
- Information: You must anchor in position 1.5 miles south from Tainio racon.
- Okay, I will anchor at given position.
- Call Kotka VTS after you have anchored.
- Okay, I will call Kotka VTS after I have anchored.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 5, Exercise 2

- Kotka VTS, Marina. **We/we/I/i** have engine problems and **I/i** am manoeuvring with difficulty.
- Marina, Kotka VTS. Do **you** need assistance?
- Negative, no assistance needed.
- Understood. Keep **me** updated about **your** situation.
- Kotka VTS, this is Nordic Star.
- Nordic Star, Kotka VTS.
- Intention: Nordic Star is ready to depart from berth number C5 to sea. Can **I/i/we** have clearance now?
- Negative. Do not proceed. Traffic information: Tanker Paula is just departing from oil terminal. Call VTS again after Paula has passed you.
- Well understood. **We/we/I** do not have permission to depart. **We/we/I** will stay alongside. **I/we/We** will call **you** again when tanker Paula has passed **me/us**.
- Friedrich Russ, Kotka VTS.
- Kotka VTS, this is Friedrich Russ.
- Information: Outbound vessel Marina in the vicinity of buoy number 10 has engine problems and **she** is manoeuvring with difficulty. Advice: Keep minimum passing distance of five cables when passing **her**.
- Well understood. **I/i/We/we** will keep clear of Marina.
- **I/i** still have problems with engines. Question: Where can **I/i** anchor?
- Question: What is **your** maximum draught/maximum draft?
- Answer. **My/my** maximum draught is six decimal five metres.
- Information: **You/you** must anchor in position 1.5 miles south from Tainio racon.
- Okay, **I/i** will anchor at given position.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 5, Exercise 3

make	speed
stay	alongside
need	assistance
have	problems
depart	from berth
keep	clear
manoeuvre	with difficulty
maintain	distance

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 5, Exercise 4

1. **Intention/intention:** I/i will reduce my speed.
2. **Advice/advice:** Turn to port.
3. **Warning/warning:** You/you are running into danger.
4. **Request/request:** I/i require a tug.
5. **Instruction/instruction:** Do/do not cross the fairway.
6. **Information/information:** Icing is expected in the Gulf of Finland.

Vessel Traffic Services (VTS) Part 2: VTS in practice, Dialogue 5, Exercise 5

Distance between the vessels is one decimal five miles.

Dredging operations south of Lonely Island. Wide berth requested.

One inbound vessel, name Rita.

I will alter course to port.

I require navigational assistance.